Software Engineering Project May-2026

Milestone 2 Deliverables for ZenStudy

Submitted By

Team Name: DeerInTheCode

Team Code : Team-051



IITM Online BS Degree Program,
Indian Institute of Technology, Madras,
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Community Services Platform for Library Management

Project Details

Software Name: ZenStudy

Department Context: Library Management System

System Type: Small Business Logistics & Community Services Platform

Date of Submission: 23-07-2026

Team Details

Team Name: DeerInTheCode
Team Code: Team-051

Name	Roll Number	Role
DIKSHA SHARMA	23f2004899	Product Manager
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GAURAV TOMAR	23f1000805	Backend Developer
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DESIGN OF COMPONENTS

1. Student Component (Booking, Membership & Self-Service APIs)

- Securely log in to the study library portal and manage account credentials.
- View membership details, membership validity, and subscribed membership plan.
- Update personal profile and contact information.
- View real-time seat and study room availability before visiting the library.
- Reserve study seats or study rooms in advance.
- Modify or cancel bookings within the permitted time period.
- View current booking details and assigned study seat.
- Present a unique check-in code to library staff during check-in.
- Check out after completing the study session.
- View booking history, attendance history, and payment history from a unified dashboard.
- Raise complaints or support requests regarding library services.

2. Library Staff Component (Operations, Attendance & Payment APIs)

- Register new students and verify membership eligibility.
- View daily bookings, expected arrivals, and current occupancy.
- Verify student bookings using unique check-in codes.
- Perform student check-ins and check-outs.
- Handle same-day walk-in admissions by allocating available seats.
- Review booking extension requests and extend bookings or allocate alternate seats based on availability.
- Mark no-show students after the configured waiting period and release reserved seats.
- Record offline payments using cash, UPI, card, or other supported payment methods.
- Manage booking approvals and booking cancellations when necessary.
- View and resolve student complaints through a centralized complaint management system.
- Monitor daily attendance and seat occupancy to support smooth library operations.

3. Admin Component (Administration, Configuration & Analytics APIs)

- Create, update, and deactivate library staff accounts.
- Approve, suspend, or manage student memberships.
- Configure study library operating hours.
- Configure booking policies, booking limits, and available seating capacity.
- Create, modify, and manage membership plans.
- Publish announcements and notifications for students and staff.
- View dashboards displaying occupancy statistics, peak usage hours, memberships, attendance, and revenue.
- Monitor overall library performance through analytical reports.
- Manage system-wide configurations and operational policies.

4. Booking & Seat Allocation Component (Reservation Management APIs)

- Maintain real-time seat availability across all study areas.
- Allocate seats during advance bookings.
- Allocate seats for walk-in students.
- Handle booking modifications and cancellations.
- Process booking extension requests.
- Reassign seats when the originally assigned seat is unavailable.
- Automatically release seats after booking cancellations or no-shows.
- Prevent double-booking through availability validation.

5. Attendance Management Component (Check-in & Occupancy APIs)

- Generate unique booking check-in codes.
- Validate student check-in codes during arrival.
- Record check-in and check-out timestamps.
- Update seat occupancy status in real time.
- Identify no-shows after the configured grace period.
- Release reserved seats for reallocation.
- Maintain attendance history for every student.

6. Payment Management Component (Payment & Transaction APIs)

- Record offline membership and booking payments.
- Support multiple payment modes (Cash, UPI, Card).

- Maintain payment transaction history.
- Link payments with memberships and bookings.
- Generate payment receipts.
- Track payment records for administrative reporting.

7. Complaint & Communication Component (Support APIs)

- Allow students to submit complaints and support requests.
- Enable staff to review, update, and resolve complaints.
- Maintain complaint history and resolution status.
- Publish announcements and notifications to all users.
- Notify students about membership expiry, booking confirmations, booking updates, and important announcements.

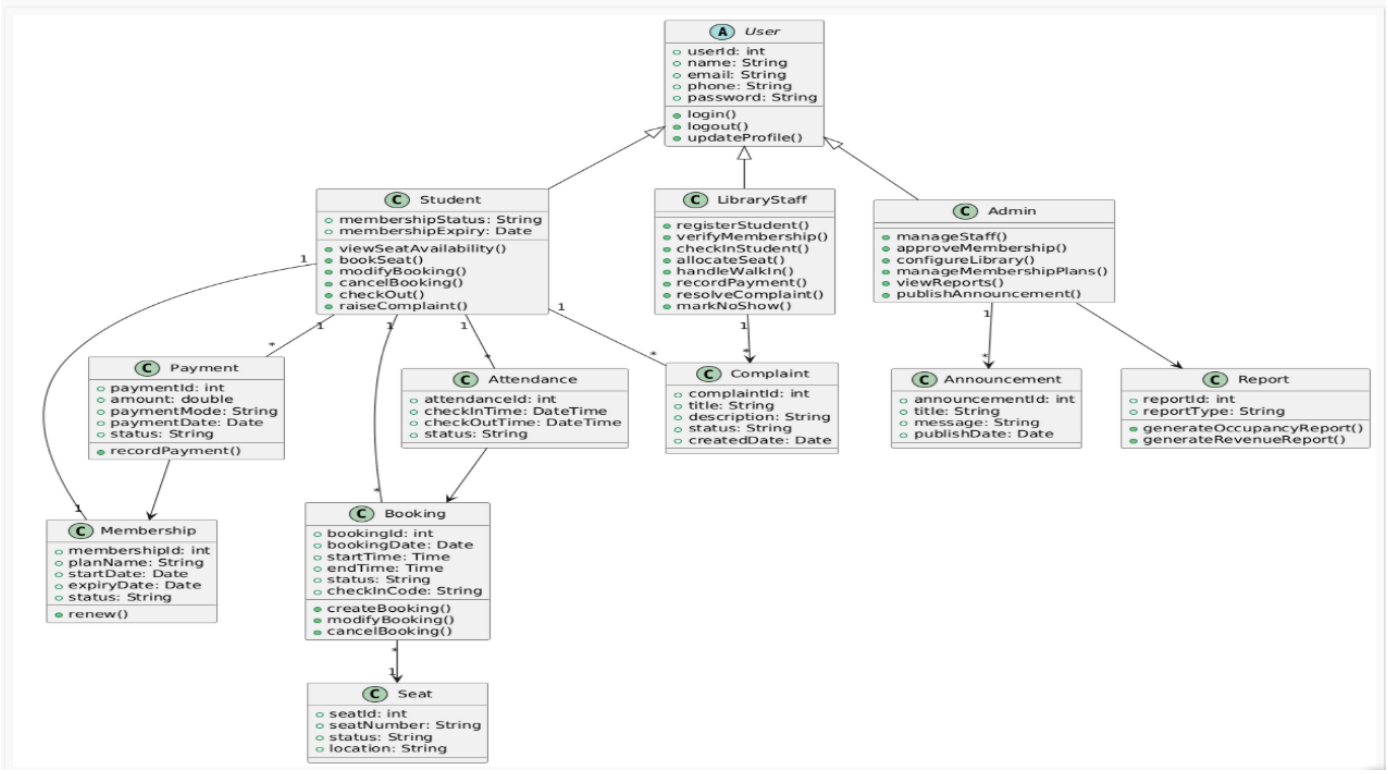
SYSTEM DESIGN- CLASS DIAGRAM

Overview

The UML Class Diagram represents the structure of the Study Library Management System (SLMS) by illustrating the key classes, their attributes, methods, and relationships. It models the interactions between Students, Library Staff, and Admin, along with core entities such as Membership, Booking, Seat, Attendance, Payment, Complaint, Announcement, and Report, providing a clear blueprint for the system's design.

Key Highlights

- User inheritance with a common User superclass extended by Student, LibraryStaff, and Admin.
- Student manages memberships, bookings, attendance, payments, and complaints.
- Library Staff handles student registration, check-ins, seat allocation, payments, and complaint resolution.
- Admin manages staff, memberships, library configurations, announcements, and reports.
- Booking is associated with Seat and Attendance to manage reservations and study sessions.
- Payment, Complaint, Announcement, and Report support financial management, issue tracking, communication, and analytics.



DATABASE DESIGN - ER DIAGRAM



Overview

The Entity Relationship (ER) diagram illustrates the database structure of the Study Library Management System by defining the key entities, their attributes, and the relationships between them. It models the core functionalities of the system, including user management, memberships, seat bookings, attendance tracking, payments, complaints, announcements, and library settings. The database is designed using a relational model to ensure data consistency, minimize redundancy, and efficiently support the day-to-day operations of the study library.

Key Highlights

- Role-based user management through the Users table, supporting Admin, Library Staff, and Students.
- Student and Membership Management with separate tables for student details, membership plans, and active memberships.
- Booking and Seat Management using the Bookings and Seats tables to manage seat allocation, reservations, and booking status.
- Attendance Tracking by recording student check-in and check-out details linked to bookings.
- Extension Requests to allow students to request additional study time, subject to staff approval.
- Payment Management for recording offline payments, payment modes, and membership transactions.
- Complaint Management to track student issues and their resolution by library staff.
- Library Configuration through the Library Settings table, enabling management of operating hours, booking limits, cancellation policies, and no-show rules.
- Announcements to facilitate communication between administrators and users.
- Well-defined relationships and foreign keys ensure referential integrity and maintain consistency across all system modules.

SPRINT SCHEDULE

Sprint 1: Project Initiation & Planning

- **Duration: 18/06/2026 – 21/06/2026**
- Finalize project topic and scope
- Identify primary stakeholders
- Finalize user research strategy
- Prepare interview questionnaire and observation plan
- Assign team roles and responsibilities
- Plan project milestones and timeline

Sprint 2: User Research & Requirements Engineering

- **Duration: 22/06/2026 – 28/06/2026**
- Conduct stakeholder interviews and observations
- Analyze interview responses and identify pain points
- Identify functional and non-functional requirements
- Define system features
- Create product backlog
- Write SMART user stories and epics
- Validate requirements with stakeholders
- Prepare Milestone 1 deliverables (Problem Statement, Users, Pain Points, User Stories)

Sprint 3: System Planning & Design

- **Duration: 29/06/2026 – 05/07/2026**
- Finalize project branding and technology stack
- Create project schedule and Gantt chart
- Set up Jira project and task assignments
- Design system architecture
- Create Component Diagram
- Create UML Class Diagram
- Design database schema (ER Diagram)
- Develop UI wireframes and page navigation
- Prepare design documentation

Sprint 4: Frontend Development (Phase I)

- **Duration: 06/07/2026 – 14/07/2026**
- Develop authentication and user management pages
- Develop membership management pages
- Develop admin dashboard and configuration pages
- Develop responsive UI components
- Perform frontend unit testing

Sprint 5: Frontend Development (Phase II)

- **Duration: 15/07/2026 – 23/07/2026**
- Develop booking and seat management pages
- Develop check-in and attendance pages
- Develop payment management pages
- Develop complaints and communication pages
- Integrate all frontend pages
- **Complete all frontend screens by 23/07/2026**

Sprint 6: Backend Development & Integration

- **Duration: 24/07/2026 – 03/08/2026**
- Develop backend APIs for all modules
- Implement business logic and database integration
- Integrate frontend with backend APIs
- Implement authentication and role-based access
- Perform API testing and fix integration issues

Sprint 7: Testing & Quality Assurance

- **Duration: 04/08/2026 – 10/08/2026**
- Execute unit, integration, and system testing
- Perform bug fixing and regression testing
- Validate user stories and acceptance criteria
- Conduct User Acceptance Testing (UAT)
- Optimize performance and usability

Sprint 8: Final Documentation & Project Submission

- **Duration: 11/08/2026 – 15/08/2026**

- Finalize project documentation
- Update README and deployment guide
- Prepare final project report and presentation
- Perform final system validation
- Deploy the application
- Submit all project deliverables

SCRUM MEETINGS SCHEDULE AND MINUTES

SCRUM Meetings: Conducted on scheduled dates via Google Meet (evening sessions)

Minutes from June 17, 2026 SCRUM Meeting:

- Conducted team introductions and discussed individual roles and responsibilities.
- Reviewed the Software Engineering project guidelines and Milestone 1 expectations.
- Evaluated the available problem statements and finalized the Study Library Management System as the project domain.
- Planned stakeholder interviews and user research activities.

Minutes from June 19, 2026 SCRUM Meeting:

- Confirmed the selected business as a Study Library after initial discussions with the owner.
- Reviewed the study library workflows and distinguished them from traditional library management systems.
- Identified the preliminary user groups and planned interviews with stakeholders.
- Finalized the interview and observation strategy.

Minutes from June 23, 2026 SCRUM Meeting:

- Reviewed interview findings and observation notes collected from stakeholders.
- Identified and documented the key pain points of students, staff and library owners.
- Finalized the user hierarchy and initiated requirement analysis.
- Began drafting SMART user stories based on the identified requirements.

Minutes from June 24, 2026 Client/TA Meeting:

- Presented the selected problem statement, user research findings and proposed dashboards.
- Explained the identified pain points and planned system features.
- Received feedback to conduct additional interviews and prioritize realistic user stories.
- Discussed the use of Jira, GitHub workflow and professional project management practices.

Minutes from June 27, 2026 SCRUM Meeting:

- Reviewed and finalized all Milestone 1 deliverables.
- Verified the project requirements and SMART user stories before submission.
- Initiated discussions on Milestone 2 planning and design activities.
- Discussed the initial task allocation for the upcoming milestone.

Minutes from July 3, 2026 SCRUM Meeting:

- Prepared the overall project schedule and sprint plan.
- Discussed the major system components and module responsibilities.
- Began designing the database schema and class diagram.
- Planned the frontend page structure and navigation flow.

Minutes from July 6, 2026 SCRUM Meeting:

- Reviewed the database schema and software design diagrams.
- Finalized the core components and their relationships.
- Discussed UI layouts for different user roles.
- Updated the project schedule and design documentation.

Minutes from July 9, 2026 Client/TA Meeting:

- Presented the current project progress and demonstrated the Jira board.
- Explained the database schema and GitHub repository structure.
- Received feedback to prioritize frontend development for the Milestone 2 submission.
- Agreed to continue backend development alongside the frontend while focusing primarily on completing the UI.

Minutes from July 15, 2026 SCRUM Meeting:

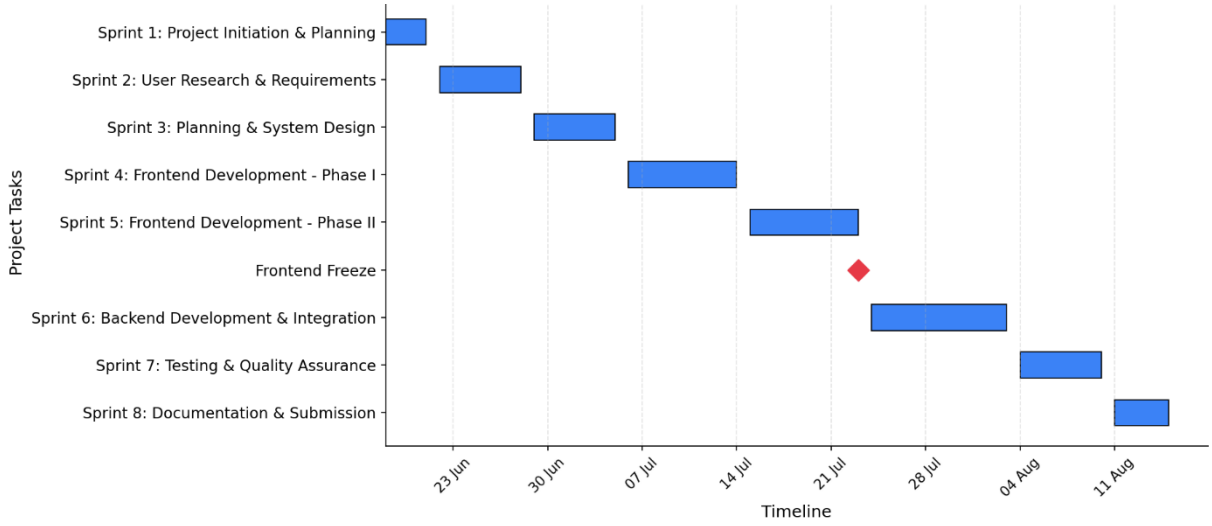
- Reviewed the implementation progress of the frontend pages.
- Verified navigation between screens and overall UI consistency.
- Discussed pending frontend pages and design refinements.
- Updated documentation and tracked remaining development tasks.

Minutes from July 21, 2026 SCRUM Meeting:

- Conducted the final review of all Milestone 2 deliverables.
- Verified the frontend pages, class diagram, database schema, Gantt chart and Kanban board.
- Reviewed the documentation and README for completeness and consistency.
- Finalized the Milestone 2 report and consolidated all submission files for submission on **23 July 2026**.

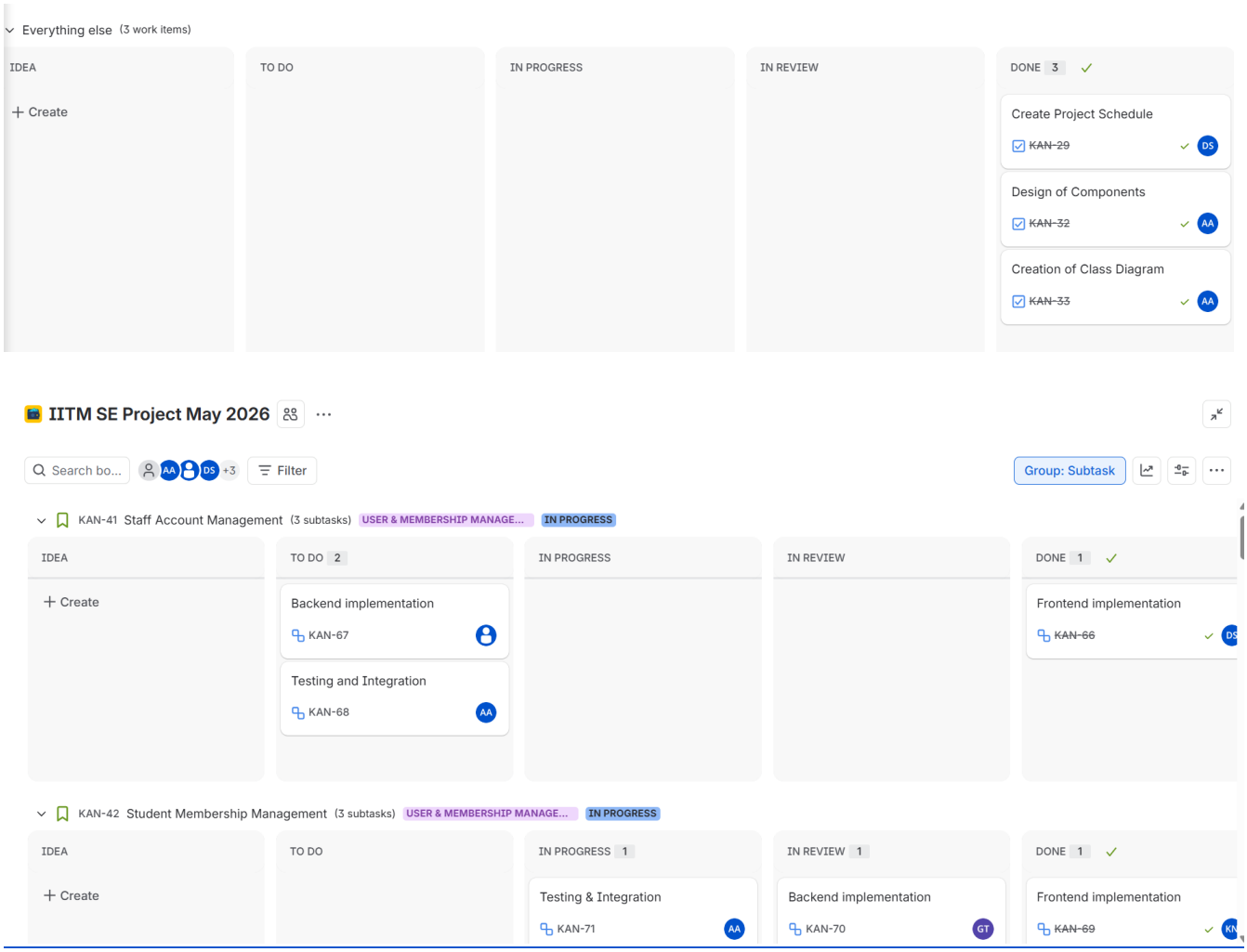
GANTT CHART

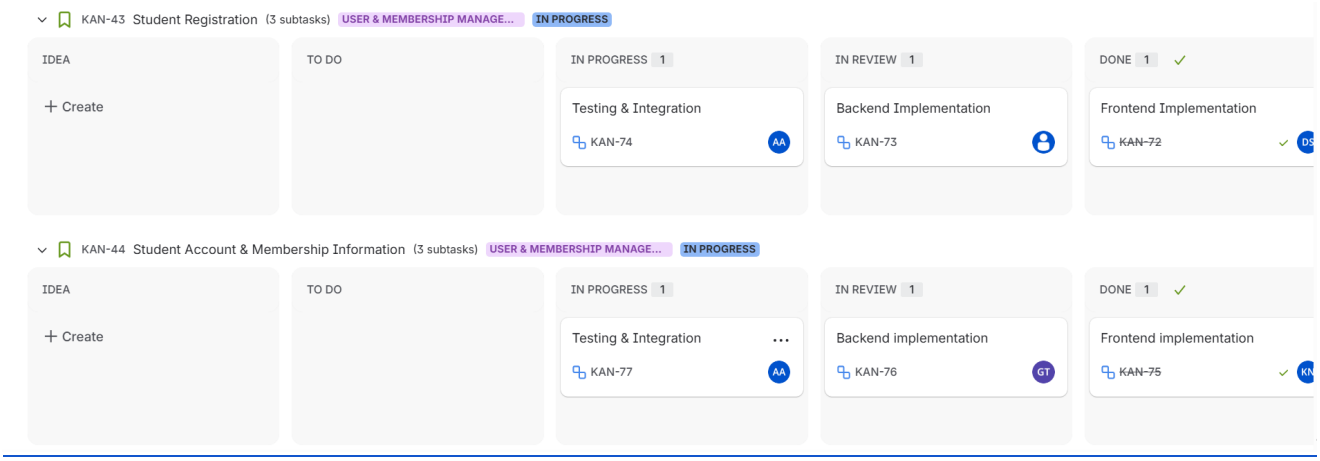
The Gantt chart illustrates the planned timeline for the Study Library Management System project, spanning from 18 June to 15 August 2026. It divides the project into eight sequential sprints covering project planning, requirements gathering, system design, frontend development, backend integration, testing, and final documentation. A milestone is included to indicate the Frontend Freeze on 23 July, ensuring all user interface development is completed before integration and testing. The schedule provides a structured roadmap for managing project activities and tracking progress throughout the software development lifecycle.

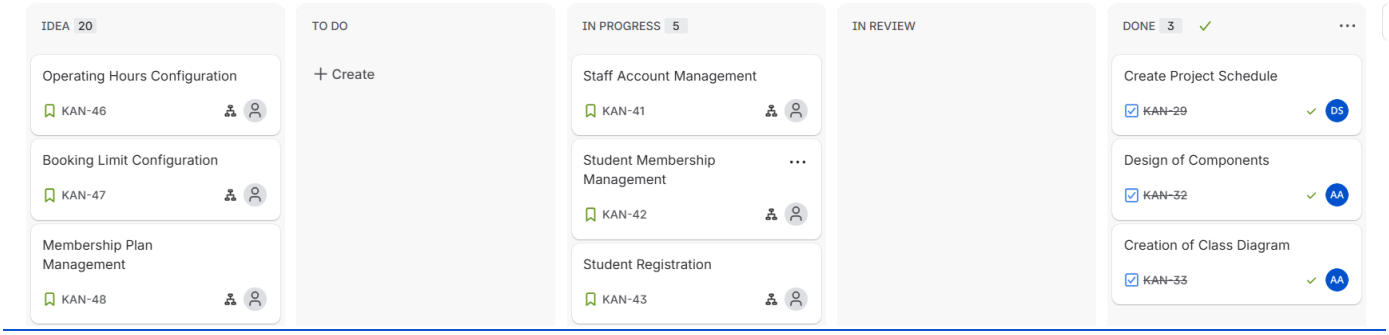


KANBAN BOARD

The Kanban board provides a visual representation of the project workflow by organizing tasks into different stages based on their current status, such as To Do, In Progress, and Done. It groups related tasks under their respective epics, enabling the team to monitor progress, prioritize work, assign responsibilities, and track the completion of project activities. This workflow-based approach improves collaboration, enhances transparency, and ensures efficient management of the Study Library Management System development process.




KAN-43 Student Registration (3 subtasks) USER & MEMBERSHIP MANAGE... IN PROGRESS
IDEA
+ Create
TO DO
IN PROGRESS 1
Testing & Integration
KAN-74
IN REVIEW 1
Backend Implementation
KAN-73
DONE 1
Frontend Implementation
KAN-72
KAN-44 Student Account & Membership Information (3 subtasks) USER & MEMBERSHIP MANAGE... IN PROGRESS
IDEA
+ Create
TO DO
IN PROGRESS 1
Testing & Integration
KAN-77
IN REVIEW 1
Backend implementation
KAN-76
DONE 1
Frontend implementation
KAN-75


IDEA 20
Operating Hours Configuration
KAN-46
Booking Limit Configuration
KAN-47
Membership Plan Management
KAN-48
TO DO
+ Create
IN PROGRESS 5
Staff Account Management
KAN-41
Student Membership Management
KAN-42
Student Registration
KAN-43
IN REVIEW
DONE 3
Create Project Schedule
KAN-29
Design of Components
KAN-32
Creation of Class Diagram
KAN-33

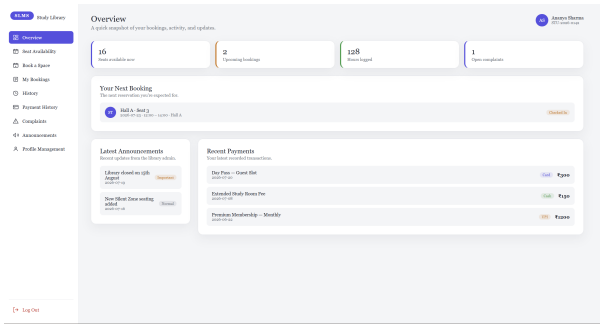
FRONTEND INTERFACE SCREENS

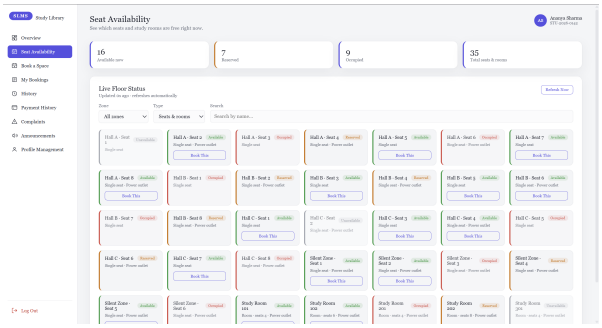
Overview

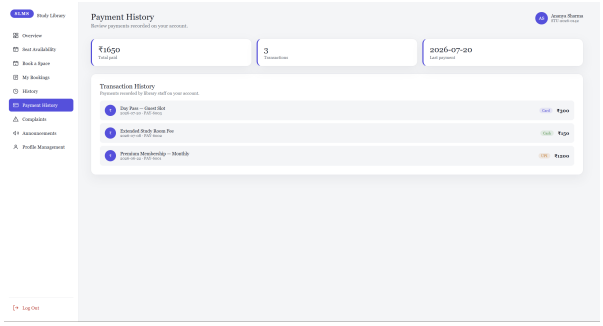
The frontend of ZenStudy is designed to support three distinct user roles — Student, Library Staff, and Admin — each with a dedicated, role-based dashboard tailored to their specific responsibilities. The interfaces are built to be intuitive, responsive, and closely aligned with the workflows and pain points identified during user research, ensuring that every user can complete their key tasks with minimal friction.

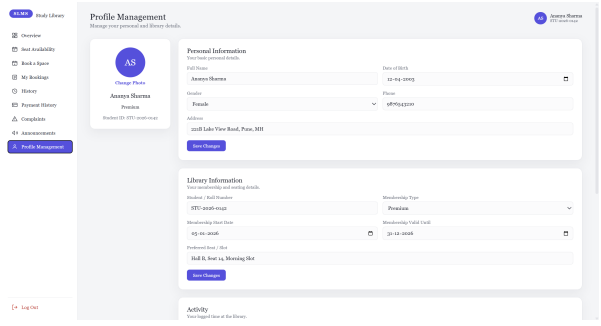
Student Dashboard

The Student dashboard allows students to log in and manage their membership, check real-time seat and study room availability, and reserve or modify bookings within the permitted time window. It shows current booking details and the assigned study seat, along with the unique check-in code used to start and end a study session. Students can also view their booking history, attendance history, and payment history, and raise complaints or support requests directly from the dashboard.





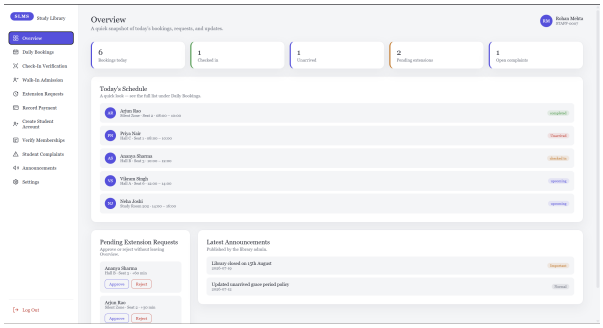
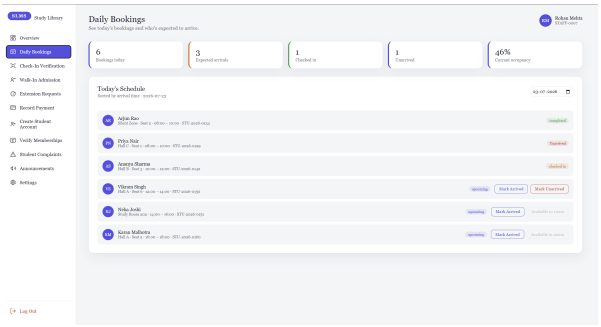
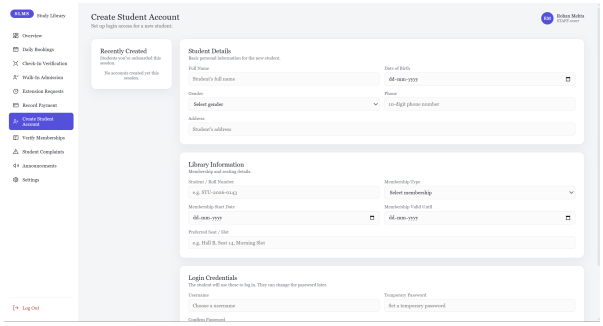
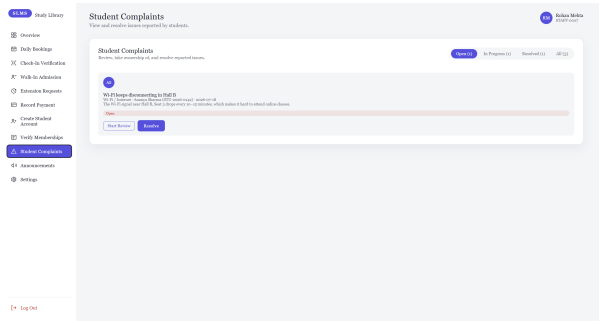




Library Staff Dashboard

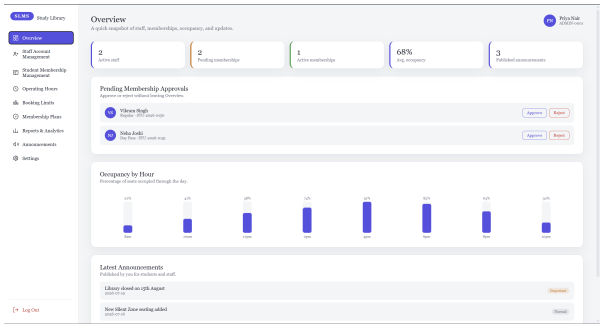
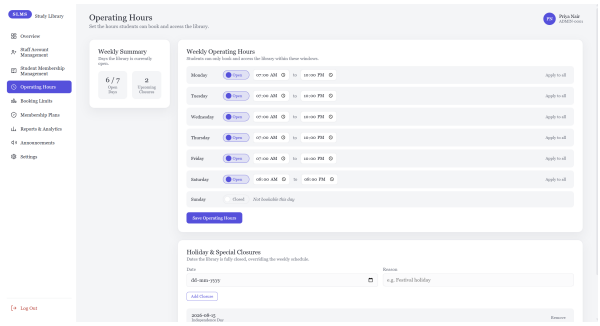
The Library Staff dashboard supports day-to-day front-desk operations, including registering new students, verifying membership eligibility, and viewing daily bookings and expected arrivals. Staff can validate student check-in codes, perform check-ins and check-outs, and handle same-day walk-in admissions by allocating available seats. The dashboard

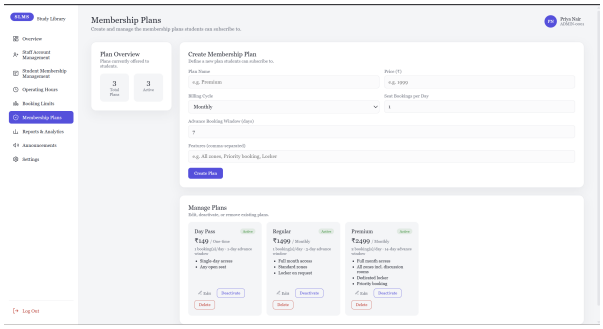
also lets staff review booking extension requests, mark no-shows after the configured waiting period, record offline payments, manage booking approvals and cancellations, and resolve student complaints through a centralized panel.

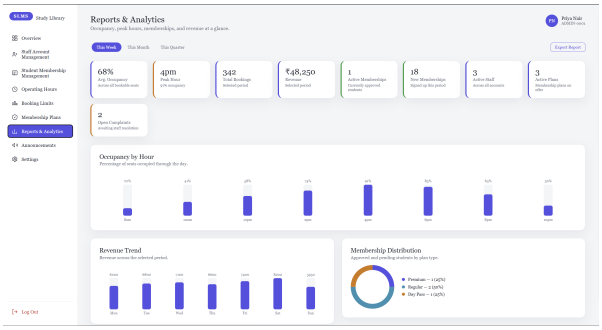





Admin Dashboard

The Admin dashboard gives administrators complete oversight and control of the platform. Admins can create, update, and deactivate library staff accounts, and approve or suspend student memberships. The dashboard supports configuring library operating hours, booking policies, and availability rules, as well as creating and managing membership plans and publishing announcements and notifications for students and staff. It also provides dashboards for occupancy statistics, payments, and overall library performance, along with system-wide configuration and operational settings.


SLMS Study Library
Overview
Staff Account Management
Student Membership Management
Operating Hours
Booking Limits
Membership Plans
Reports & Analytics
Announcements
Settings
Log Out
Membership Plans
Create and manage the membership plans students can subscribe to.
PS Priya Nair ADMIN
Plan Overview
Plans currently offered to students.
3 Total Plans
3 Active
Create Membership Plan
Define a new plan students can subscribe to.
Plan Name
e.g. Premium
Price (₹)
e.g. 1999
Billing Cycle
Monthly
Seat Bookings per Day
1
Advance Booking Window (days)
7
Features (comma-separated)
e.g. All zones, Priority booking, Locker
Create Plan
Manage Plans
Edit, deactivate, or remove existing plans.
Day Pass
Active
₹149 / One-time
1 booking/day · 1 day advance
Single-day access
Any open seat
Edit
Deactivate
Delete
Regular
Active
₹1499 / Monthly
1 booking/day · 3 day advance
Full month access
Standard zones
Locker on request
Edit
Deactivate
Delete
Premium
Active
₹2499 / Monthly
2 bookings/day · 14 day advance
Full month access
All zones incl. discussion rooms
Dedicated locker
Priority booking
Edit
Deactivate
Delete


SLMS Study Library
Overview
Staff Account Management
Student Membership Management
Operating Hours
Booking Limits
Membership Plans
Reports & Analytics
Announcements
Settings
Log Out
Reports & Analytics
Occupancy, peak hours, memberships, and revenue at a glance.
PS Priya Nair ADMIN
This Week
This Month
This Quarter
Export Report
68% Avg. Occupancy Across all bookable seats
4pm Peak Hour 87% occupancy
342 Total Bookings Selected period
₹48,250 Revenue Selected period
1 Active Memberships Currently approved students
18 New Memberships Signed up this period
3 Active Staff Across all accounts
3 Active Plans Membership plans on offer
2 Open Complaints awaiting staff resolution
Occupancy by Hour
Percentage of seats occupied through the day.
Revenue Trend
Revenue across the selected period.
Membership Distribution
Approved and pending students by plan type.
Premium — 1 (25%)
Regular — 2 (50%)
Day Pass — 1 (25%)